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2026 IEEE 5th World Conference on Applied Intelligence and Computing : Submission (2839) has been created.

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Track Name: AIC2026

Paper ID: 2839

Paper Title: Enhancing Phishing Detection Using Hybrid ML-LLM Models with SHAP Explainability for AI-Generated Attacks

Abstract:

The fast evolution of generative AI has changed how cybersecurity threats appear, especially phishing attacks. Phishing messages are generated and more personal due to current AI tools, such as ChatGPT and Grok; therefore, traditional methods used to detect these forms of attack are becoming less effective. Phishing attacks also result in global financial losses; they have increased most noticeably in developing parts of the world, like India.

Traditional methods of phishing detection using machine learning focus on features created by humans and have difficulty working well on phish that have been created or modified by other AIs or perpetrators of crime. LLM have good performance on semantic tasks, but LLM's lack of ability to show how they work makes it difficult to use them in systems that require high levels of accuracy and security. This paper proposes a hybrid phishing detection framework that integrates machine learning and LLM-based approaches with SHAP explainability. The proposed system combines URL-based feature learning using Random Forest and XGBoost with text-based contextual analysis using DistilBERT. SHAP is employed to provide transparent explanations for model predictions, enhancing trust and usability for security analysts. Experimental evaluation demonstrates that the hybrid approach significantly outperforms standalone models in terms of accuracy and robustness, particularly against AI-generated phishing attacks.

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